

Crank arm optimisation

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Table of Contents

Summary	3
Discussion	4
Verification.....	4
Problem definition and assumptions	4
Analytical calculations vs FEA.....	4
Investigation/optimisation.....	7
Objectives and constraints	7
Topology optimisation	7
Conversion into manufacturable geometry	8
Initial static study.....	10
Parametric design	14
Final static study	16
Mesh convergence	17
Buckling test.....	17
Practicality/Manufacturability.....	18
Material selection	18
Manufacturing process.....	18
Manufacturing constraints reflected in the design	18
Results	19
Conclusion.....	20

Summary

This analysis successfully optimised the geometry of a bicycle crank arm for structural performance and manufacturability while achieving a significantly reduced mass. The requirements of a maximum Von Mises stress of 220 MPa and a deflection between 1mm and 2mm. A set of hand calculations was done to test the effectiveness of the FEA software, finding the simplification of a large beam to be too robust of an option. A topology optimisation on a template part with an increased volume identified areas of high importance in principal stress paths that were used to create a draft design. Static studies were then run again on the draft design to confirm which loading direction held the greatest stress. A parametric design study was conducted on the size of the cut taken out of the face furthest from the face, finding an optimised cut of 10mm. Static tests were run again to confirm the maximum stress and deflection in the component. A mesh convergence plot was done to confirm the results converged within the allowable limit. A final buckling test was performed to confirm stability under compression. The material selection of aluminium 6061-T6 was justified and high pressure die casting (HPDC) was selected as the manufacturing process. The results were then compared to the original sub-optimal part, finding a mass reduction of 50.89% while remaining in the limits of stress and deflection.

Discussion

Verification

Problem definition and assumptions

- Static pedal load $F = 1000 \text{ N}$ applied at the pedal centre
- Pedal centre offset from the face of the crank by $e = 75 \text{ mm}$
- The sprocket end is fully fixed at the faces of the sprocket hole
- Distance along the beam from fixed geometry to force $L = 165 \text{ mm}$
- Material is Aluminium 6061-T6 ($E = 69 \text{ GPa}$, $\sigma_y = 275 \text{ MPa}$)
- Must remain within 80% of the yield stress ($0.8\sigma_y = 220 \text{ MPa}$)

Analytical calculations vs FEA

To show how closely analytical methods align with FEA, a simplified geometry has been analysed in both methods. This will be used to compare the results and comment on the accuracy and confidence that can be given to future FEA results. The crank arm is modelled as a cantilever rectangular beam fixed at spindle end, with the dimensions $L = 0.165\text{m}$, $b = 35 \text{ mm}$, $h = 32 \text{ mm}$.

Analytical calculations

Moment of inertia:

$$I = \frac{bh^3}{12} = \frac{0.035 \times 0.032^3}{12} = 9.6 \times 10^{-8} \text{ m}^4$$
$$J = \frac{bh(b^2 + h^2)}{12} = \frac{(0.035 \times 0.032)(0.035^2 + 0.032^2)}{12} = 2.10 \times 10^{-7} \text{ m}^4$$

The crank undergoes two stresses, one bending and one torsional:

Bending (where $y = \frac{h}{2}$):

$$M = F \times L = 1000 \times 0.17 = 165 \text{ Nm}$$

$$\sigma = \frac{My}{I} = \frac{165 \times \left(\frac{0.032}{2}\right)}{9.6 \times 10^{-8}} = 27.5 \text{ MPa}$$

Torsional (Where $e = 0.075\text{m}$, $r_{eff} = \frac{b}{2}$ for a worst case):

$$T = F \times e = 1000 \times 0.075 = 75 \text{ Nm}$$

$$\tau = \frac{Tr}{J} = \frac{75 \times \left(\frac{0.035}{2}\right)}{2.10 \times 10^{-7}} = 6.25 \text{ MPa}$$

Deflection:

$$\delta = \frac{FL^3}{3EI} = \frac{1000 \times 0.165^3}{3 \times 69 \times 10^9 \times 9.6 \times 10^{-8}} = 0.000227m \approx 0.23mm$$

Resultant stress:

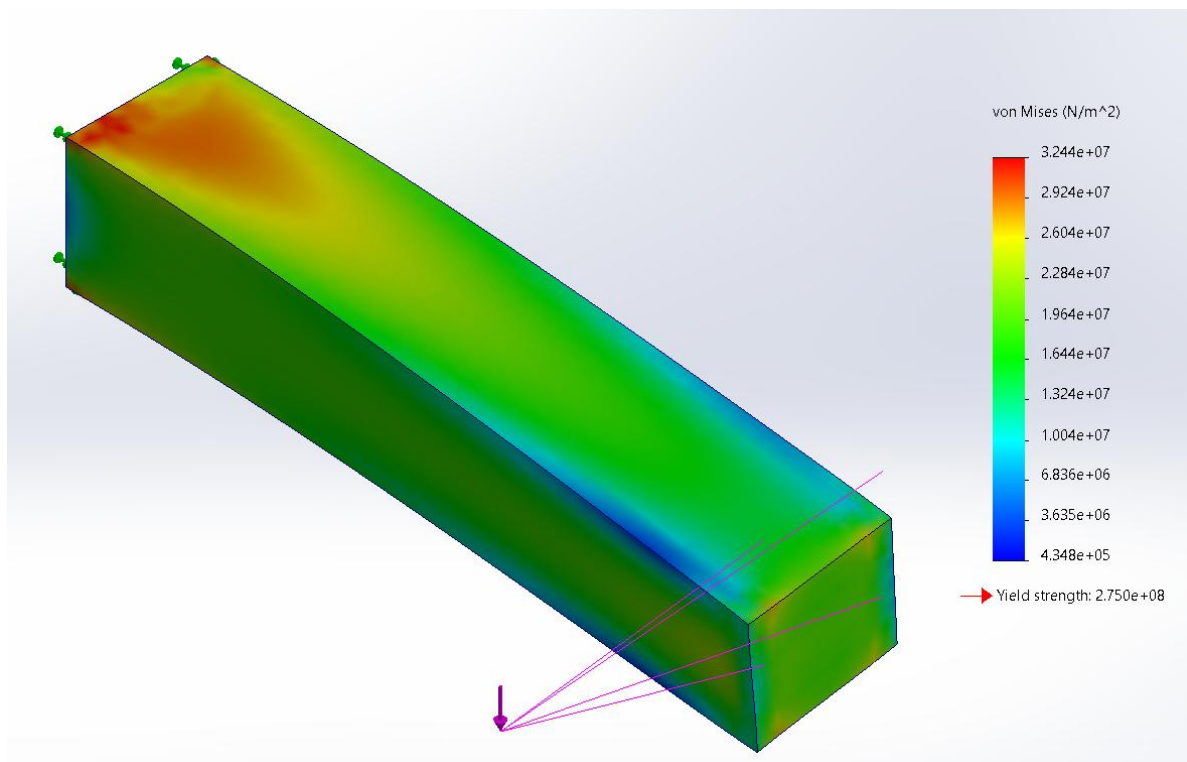
$$\sigma_{vm} = \sqrt{\sigma^2 + 3\tau^2} = \sqrt{27.5^2 + 3 \times 6.25^2} = 29.55 MPa$$

SolidWorks FEA:

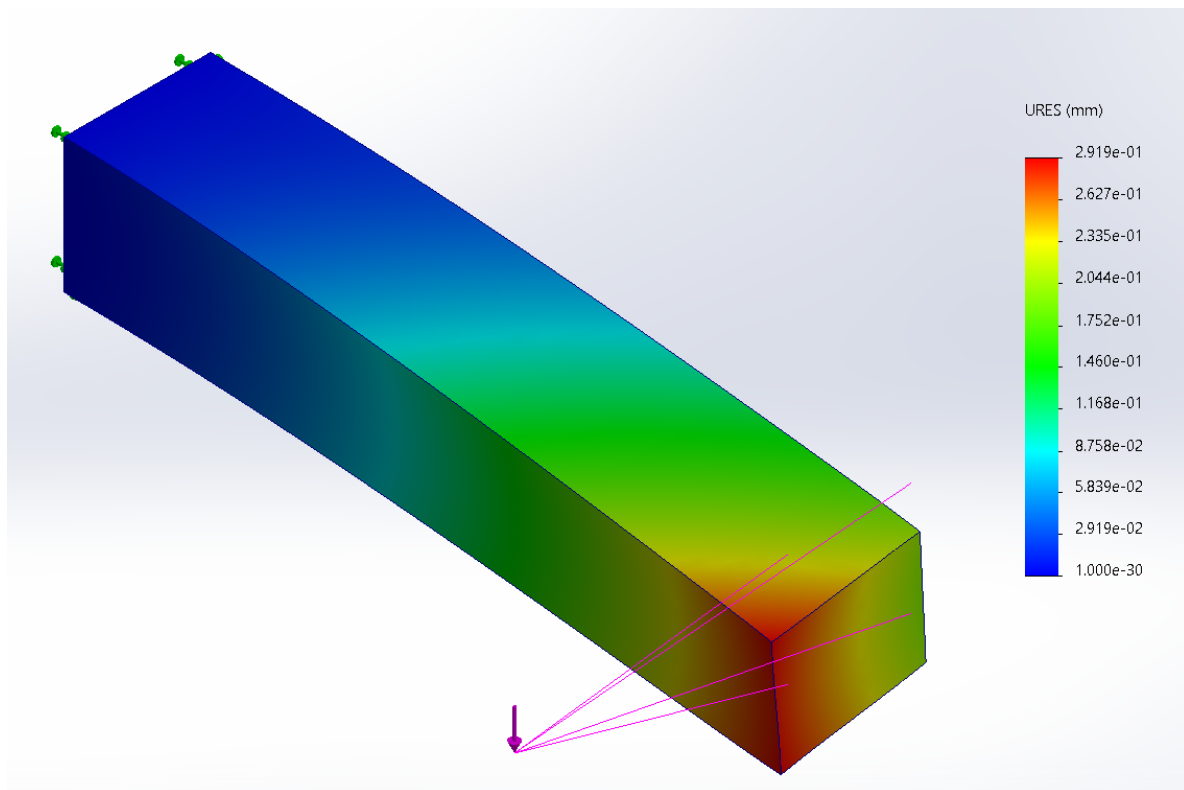
Setup:

- Geometry
 - Rectangular beam with the dimensions supplied earlier
- Material
 - Aluminium 6061-T6
 - $E = 69 GPa$
- Boundary conditions
 - Fully restrained face on the spindle end
- Load
 - A remote load of $F = 1000 N$
 - An eccentricity of $75 mm$ from the face
- Mesh
 - As fine a mesh allowed by SolidWorks (13035 nodes)

Stress results:



Displacement results:



Comparison

The results when compared show that the stress and deflection calculated by the FEA simulation are higher than the analytical results by 10% and 15% respectively. This shows how precise FEA models can be when evaluating stresses and displacements when given the same boundaries as analytical methods as it allows for a higher level of accuracy than what analytical methods provide. This shows that FEA analysis is a useful tool to use for this optimisation to take into account a more real-world take on how forces react with components.

	Analytical	FEA	Upper limit	Difference
Max σ_{vm} (MPa)	29.55	32.44	220	10%
Max URES deflection (mm)	0.23 mm	0.29 mm	2 mm	15%

Safety factor:

$$FOS = \frac{\sigma_y}{\sigma_{vm}} = \frac{220}{32.44} = 6.78$$

The results are well within the given constraints for both stress and deflection both being only 15% of their targets values and having an extremely conservative safety factor of 6.78. This means that there is a lot of room for improvement in optimisation of just a beam, saving mass and therefore cost in production.

Investigation/optimisation

Objectives and constraints

Objectives:

- Minimise the mass of the crank arm while ensuring performance in use.

Constraints:

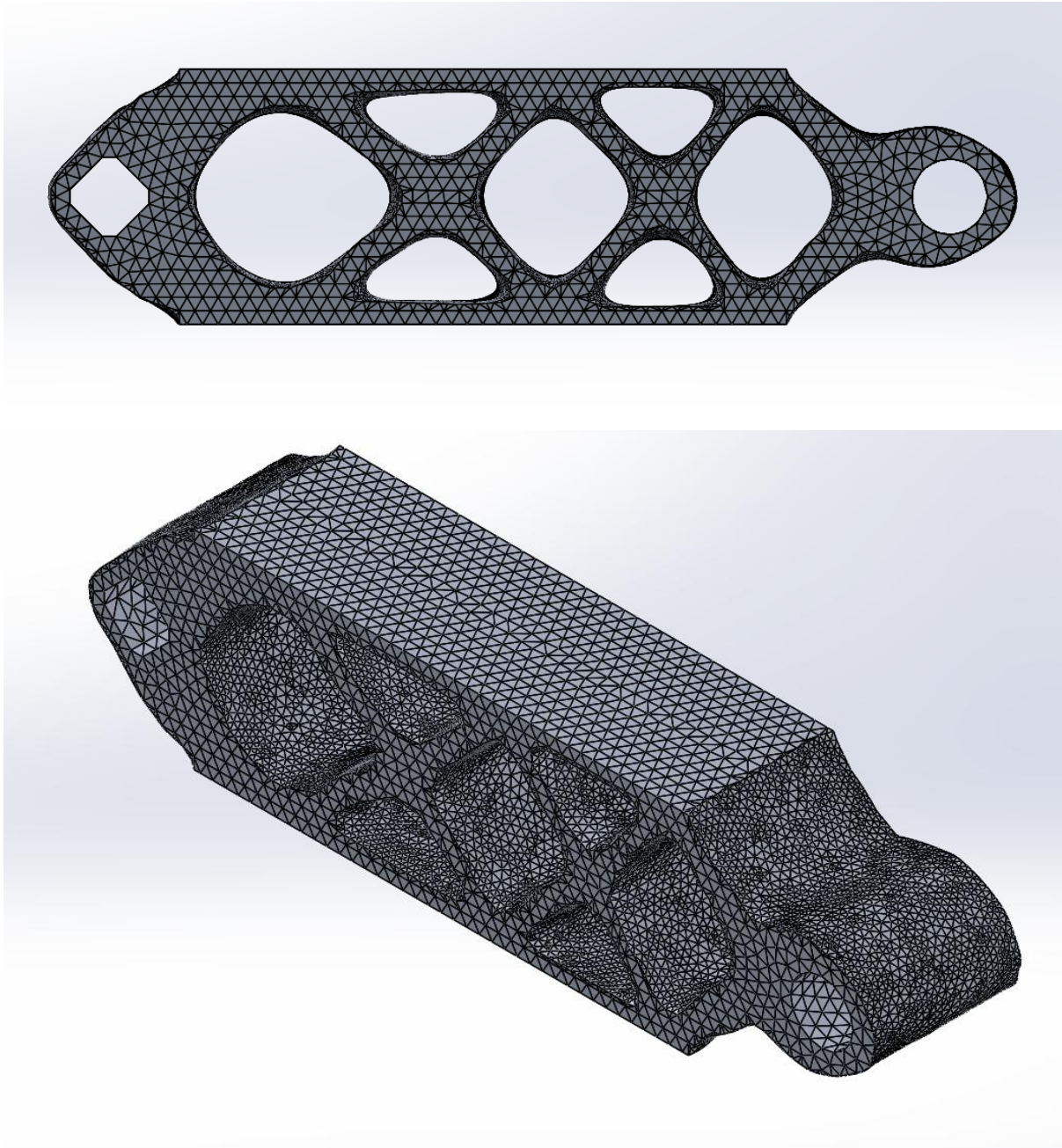
- Maximum Von Mises stress less than 200 MPa (safety factor of 1.25)
- Tip deflection between 1mm and 2mm
- Restriction to forging, casting or subtractive machining operations

Topology optimisation

Setup:

- Geometry
 - Pedal and sprocket holes left
 - 50 mm height to allow for a wider volume of mass to be placed in (improving moment of area)
 - 35 mm thickness to remove the curved length path, reducing weak points in the geometry
- Material
 - Aluminium 6061-T6 ($E=69$ GPa)
- Boundary conditions
 - The sprocket end is fully fixed at the faces of the sprocket hole
- Load
 - A remote load of $F = 1000$ N
 - An eccentricity of 75 mm from the face
- Mesh
 - A medium mesh (17719 nodes)
 - Draft quality to help with running time
- Goals and constraints
 - Mass reduction target of 60%
 - Safety factor of at least 1.25
- Manufacturing controls
 - Preserved region of 3mm around both holes
 - Symmetry control to ensure adequate structural integrity in all orientations in the pedal stroke
 - Stamping (pull direction only) De-mold control to ensure ease of manufacturing (stopping hollow centres which cannot be casted easily)

Smoothed topology results:

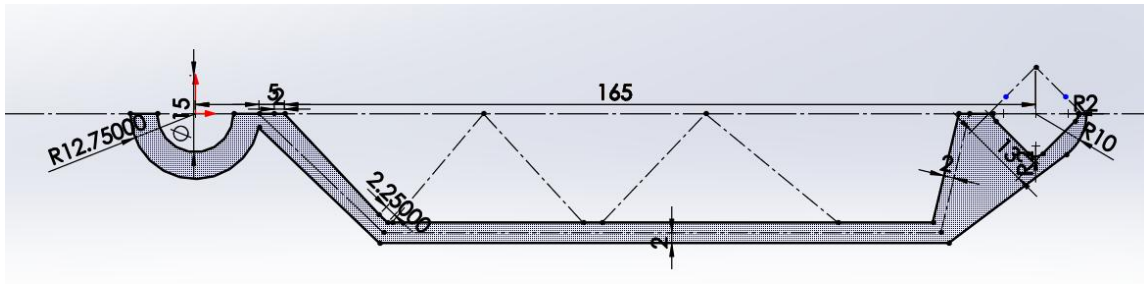


Findings:

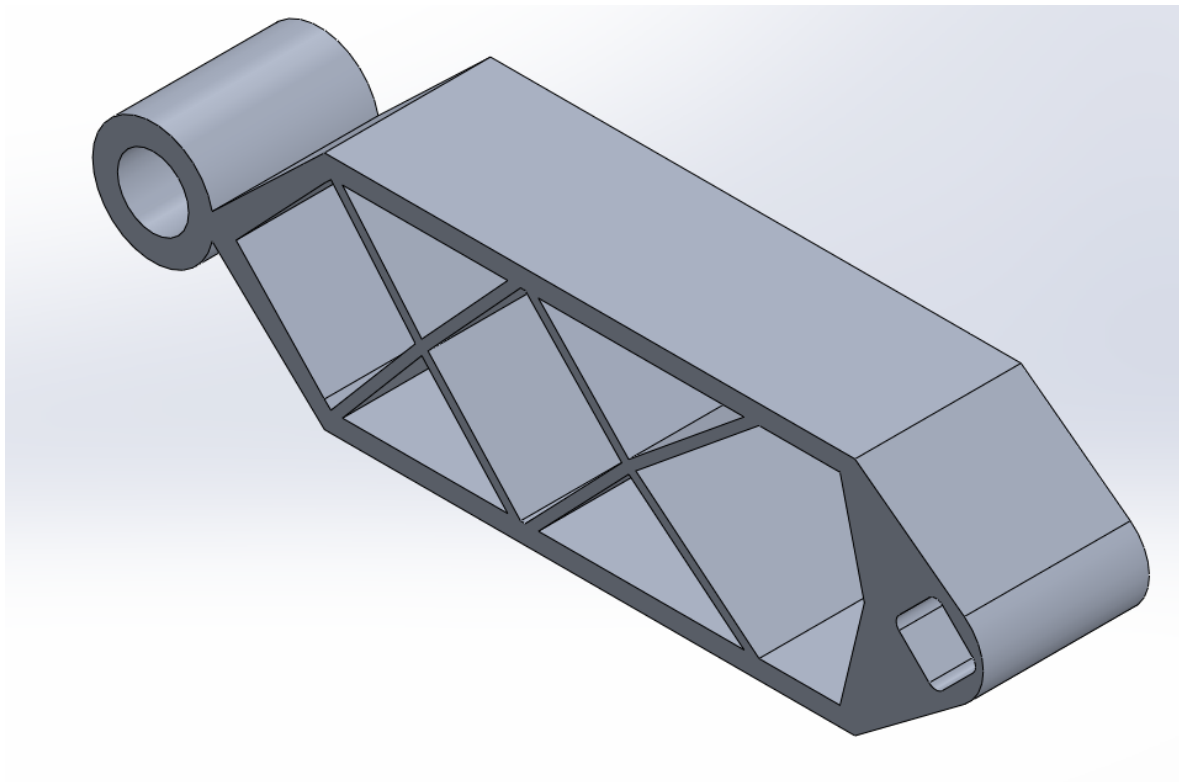
- Cross-patterned webs formed showing principal stress paths
- Not much material needed around the holes
- Focus of need for material further away from neutral axis (to increase moment of area)

Conversion into manufacturable geometry

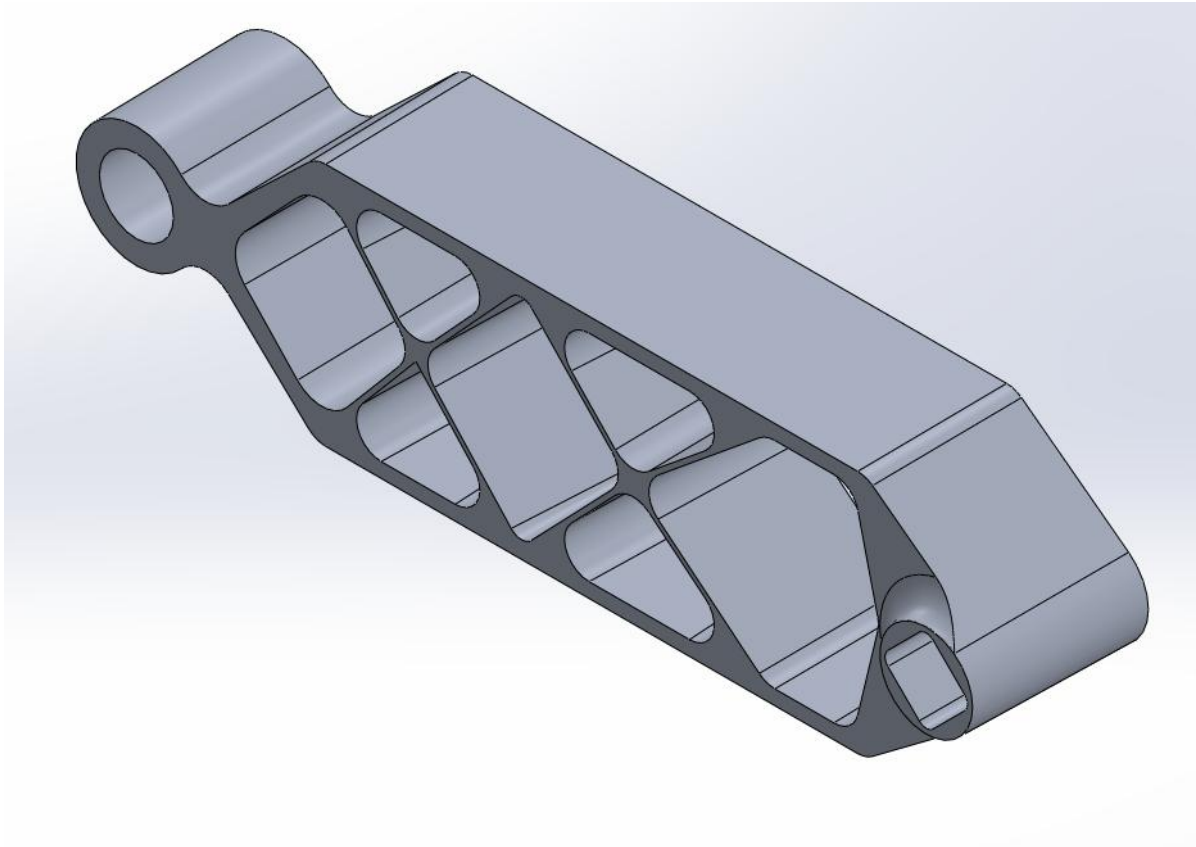
The web pattern was copied from the geometry into a sketch.



The feature was extruded by 35 mm and mirrored to achieve a symmetrical part



A cut of 5mm was taken out of the face furthest from where the force is being applied and a draft angle was applied to all relevant faces to ensure ease of ejection in casting



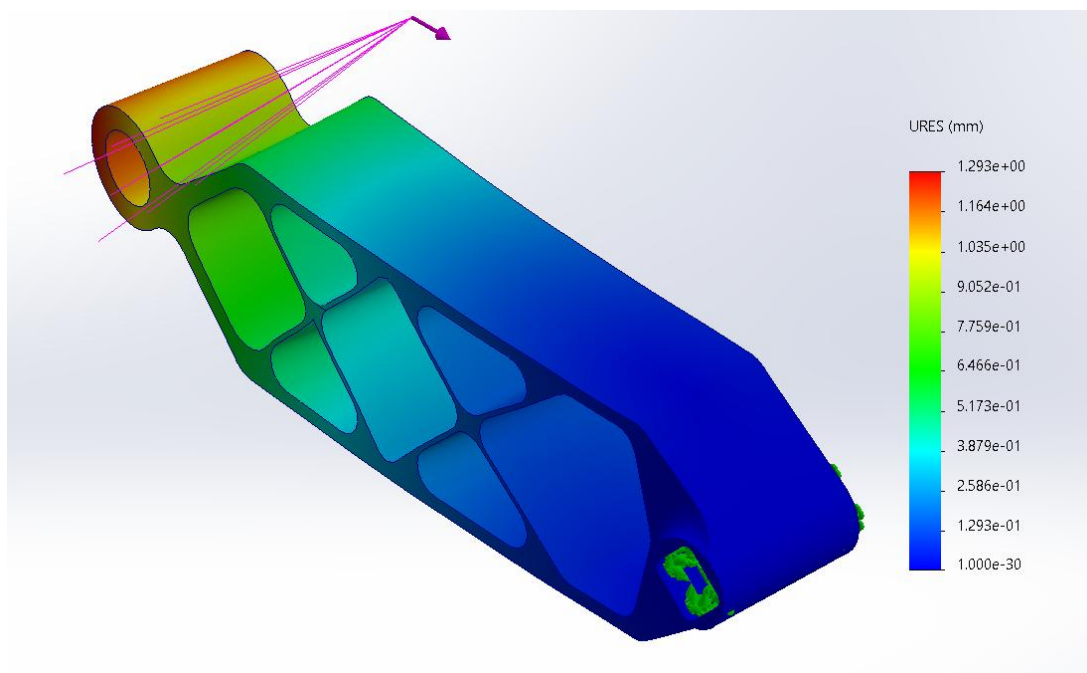
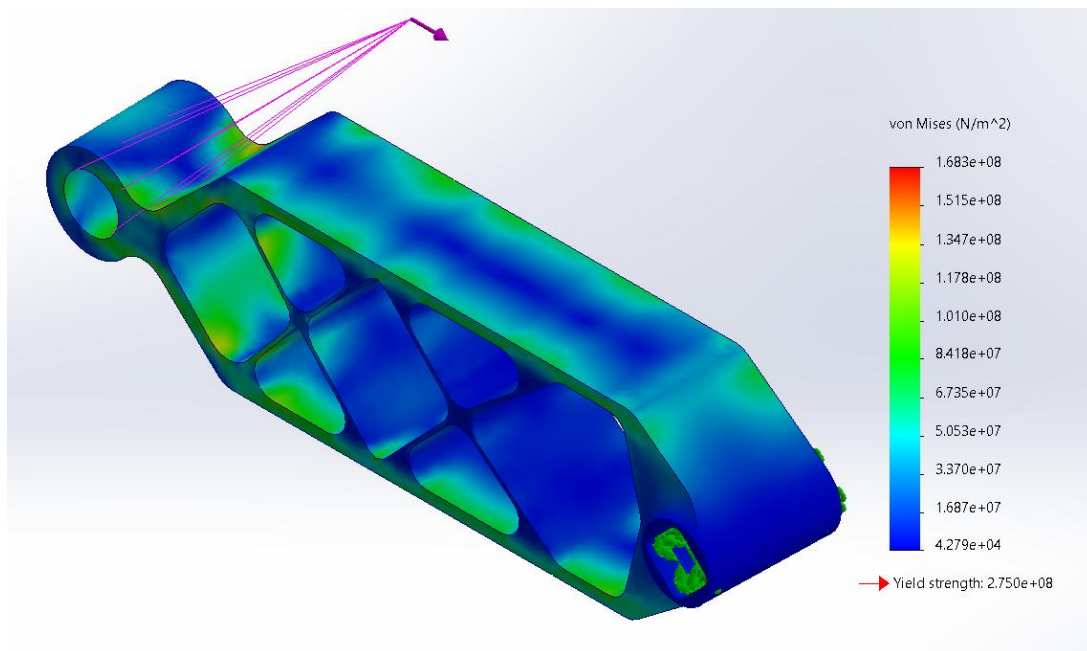
Fillets were added to all sharp edges to reduce areas of stress concentration

Initial static study

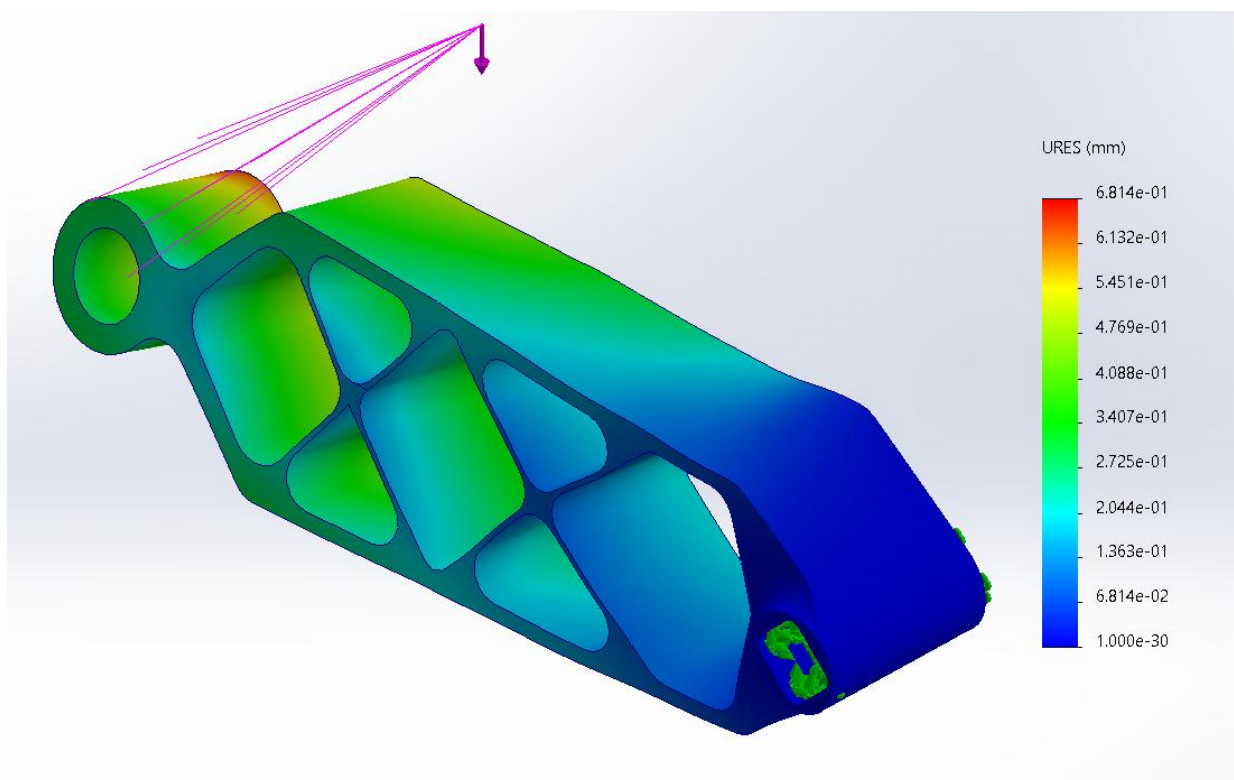
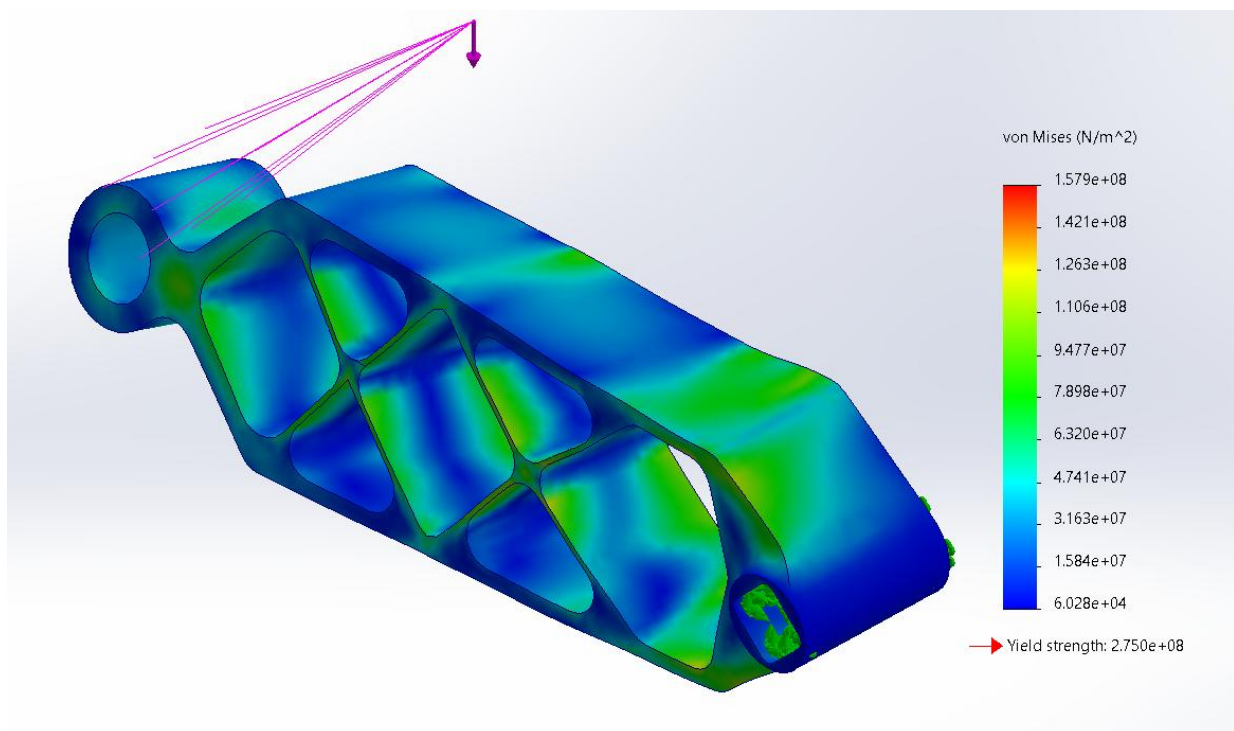
Due to a pedal stroke being 360° , studies were carried out at the extreme values to find the maximum stresses. The finest mesh available was used. The values chosen (with respect to the vertical axis) were:

- 0° – The force acting towards the sprocket end of the arm
- 90° – The force acting perpendicular to the arm's length (270° gives the same results but reversed due to symmetry along this axis)
- 180° – The force acting away from the sprocket end of the arm

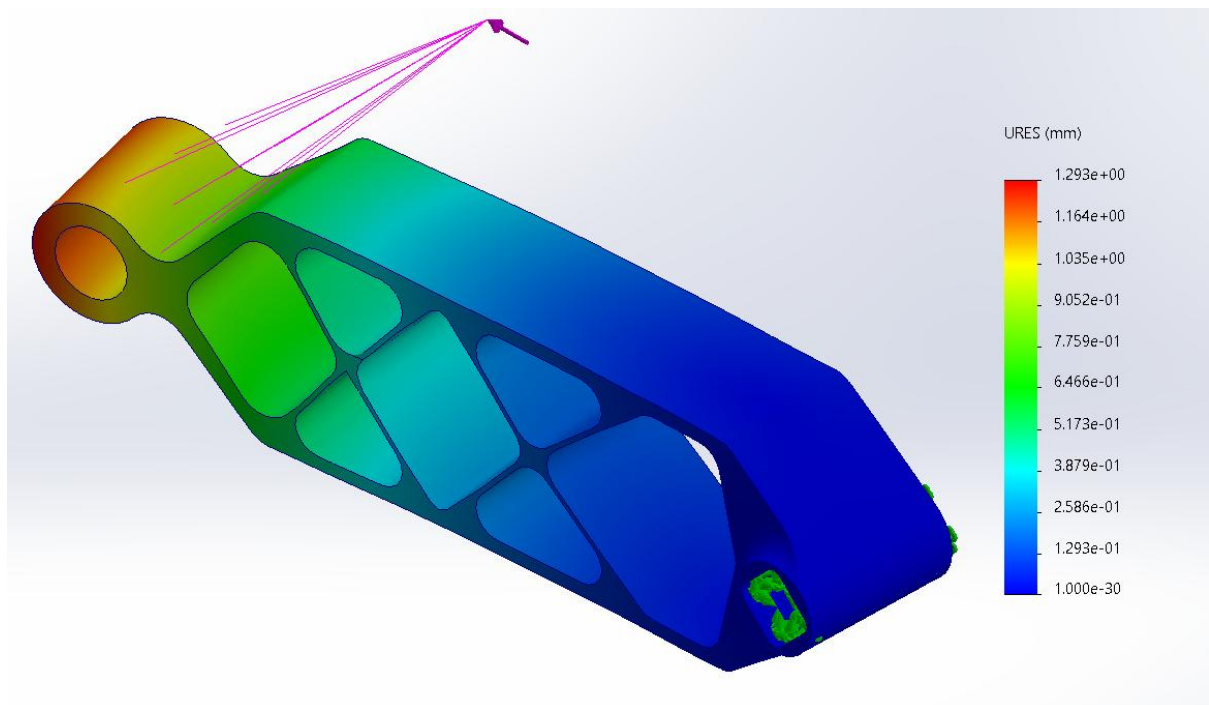
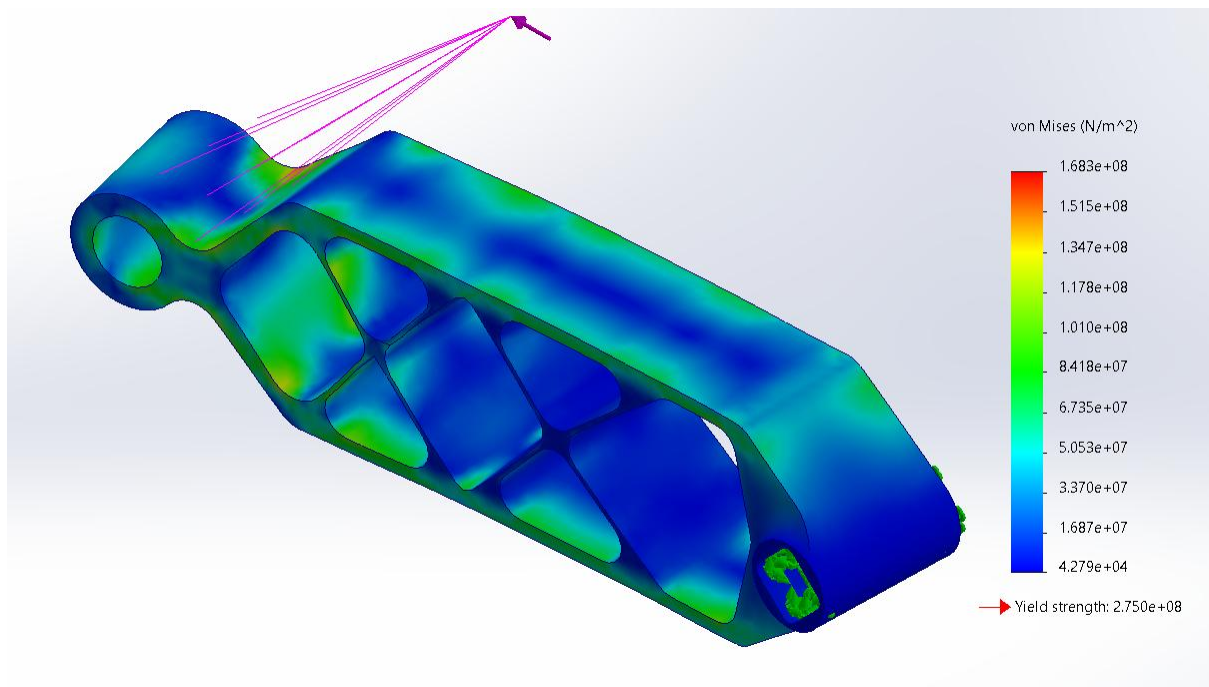
At 0°:



At 90°:



At 180°:



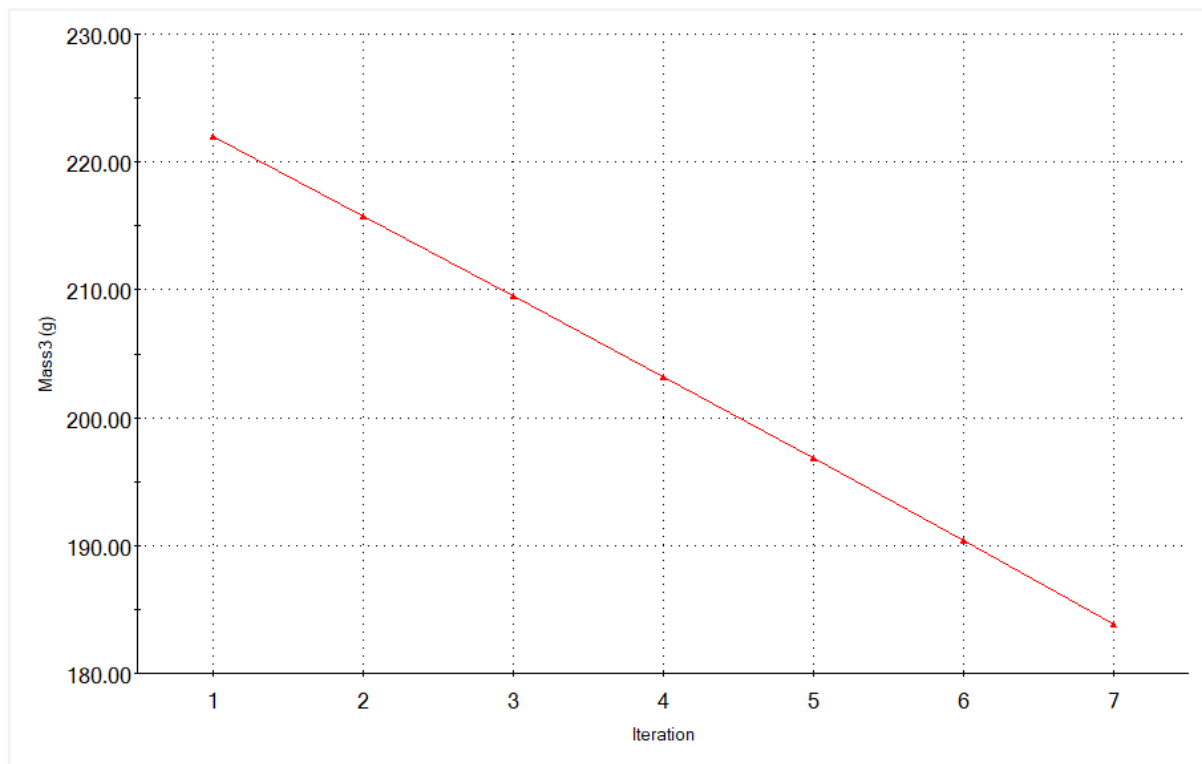
Findings:

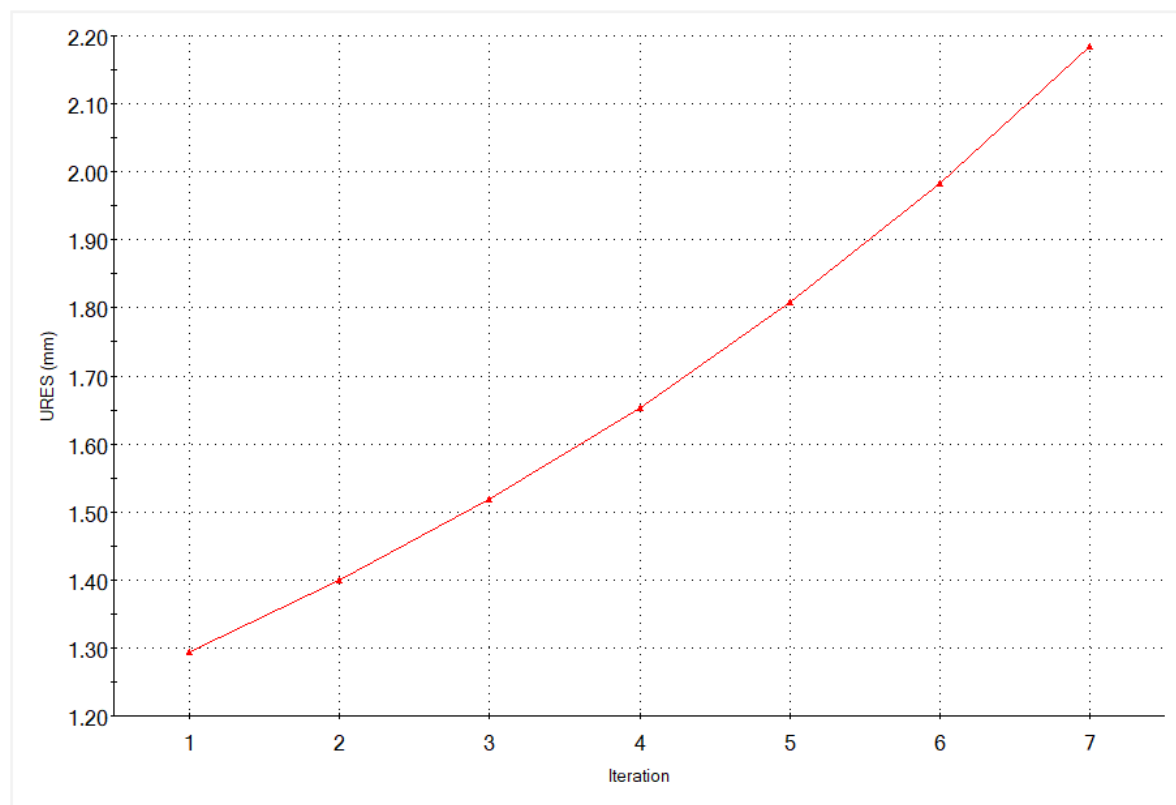
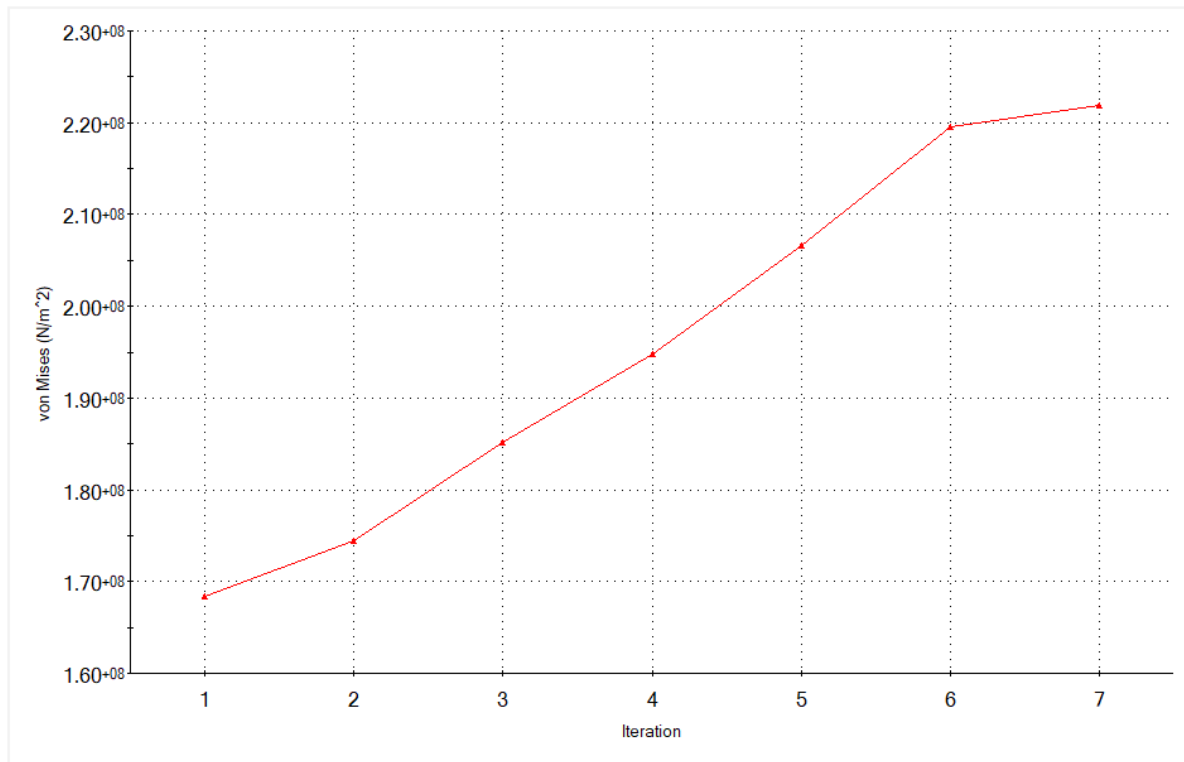
The maximum stress and deflection in the part are at 0° and 180° (which give identical results). This orientation will be used for the parametric design.

Parametric design

A design study was run on the cut taken out of the face, varying from 5mm to 11 mm in steps of 1 mm. The stress and deflection values were taken from the position of 0° from the vertical.

Iteration	Mass (g)	Max σ_{vm} (MPa)	Max URES deflection (mm)
Baseline (5 mm cut)	222.03	168.32	1.29
1 (6 mm cut)	215.78	174.46	1.40
2 (7 mm cut)	209.51	185.21	1.52
3 (8 mm cut)	203.18	194.76	1.65
4 (9 mm cut)	196.79	206.56	1.81
5 (10mm cut)	190.34	219.54	1.98
6 (11mm cut)	183.87	221.89	2.19

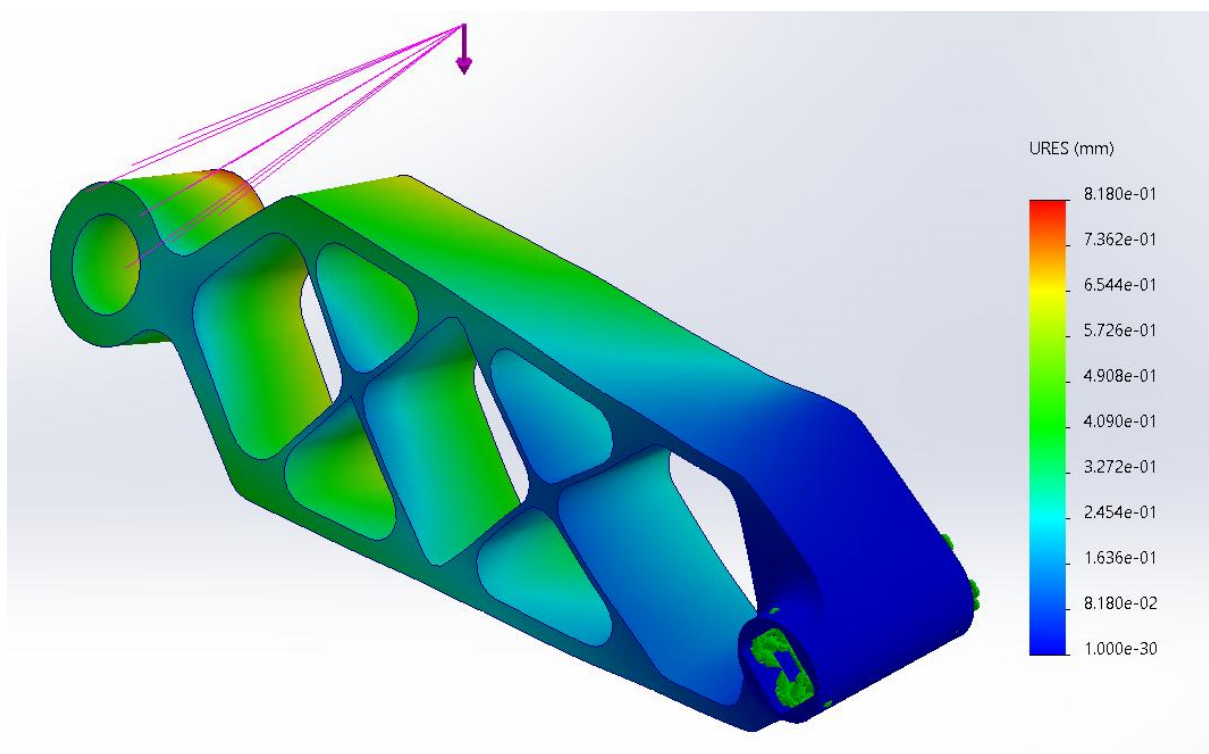
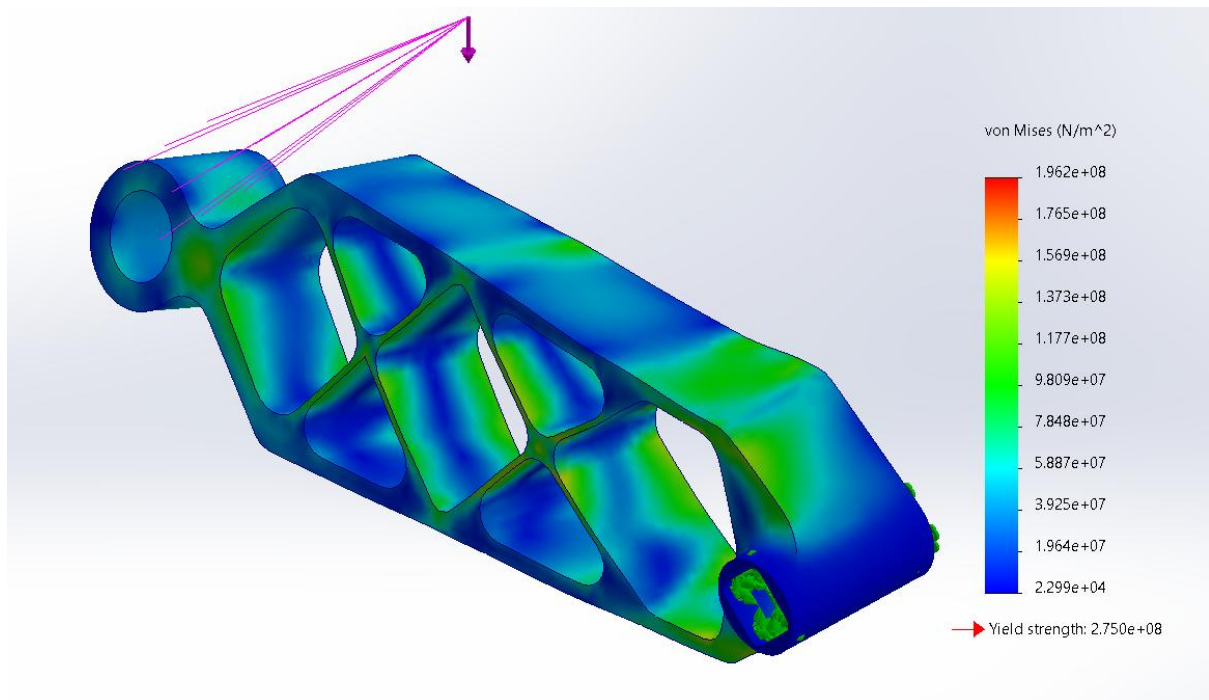




A value of 11 mm was found to be too high to stay within the limit of 220 MPa and 2 mm deflection. The value of 10 mm was chosen as the ideal cut size.

Final static study

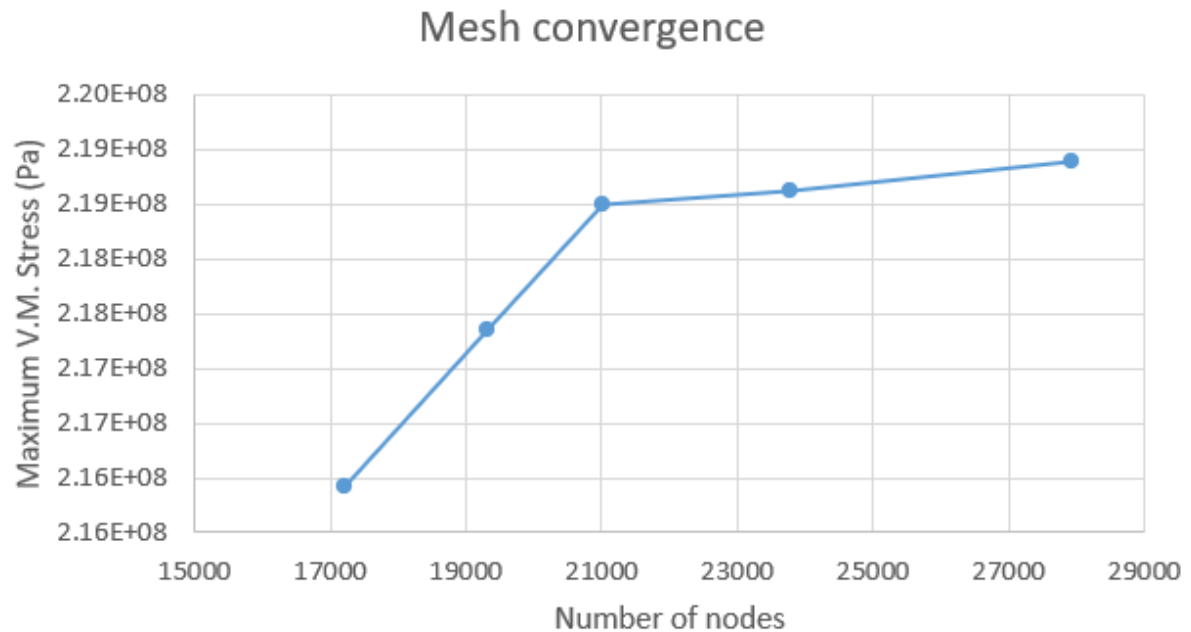
A static study was then conducted at 90° to ensure that dimension also remained in the limits in the other orientation:



The values fell well within the acceptable region, confirming the suitability of the design.

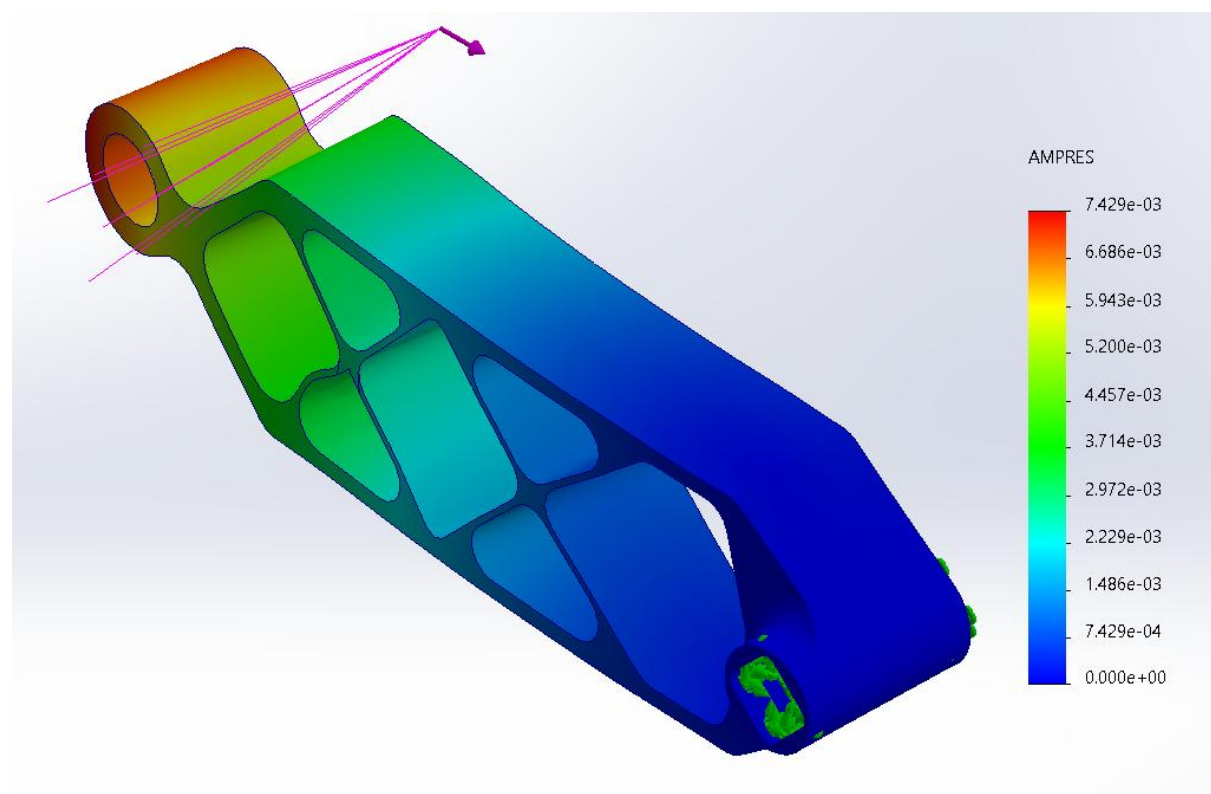
Mesh convergence

Multiple mesh sizes were used to find a convergence value at the highest stress position of 0°. This converged to less than the maximum von Mises stress, confirming the suitability of the design.



Buckling test

As buckling occurs under compressive loads, a test was ran at the position of 0°



The study showed no signs of any buckling as the resultant amplitude is low and the buckling load factor was 30.693, meaning the load applies is vastly less than what is needed to buckle the arm.

Practicality/Manufacturability

Material selection

Researching into commonly used-in-industry crank arm materials, there were two main options for the material selection, Aluminium 6061-T6 and Aluminium 7075-T6. Titanium alloys were disregarded due to them being unnecessarily strong for the requirements.

Factor	6061-T6	7075-T6
Cost	Lower	Higher
Density	2700 kgm^{-3}	2810 kgm^{-3}
Yield stress	275 MPa	503 MPa
Elastic modulus	69 GPa	71.7 GPa
Hardness	95 HB	150 HB

Aluminium 6061-T6 was chosen as the material because:

- Cheaper – 66.7% the price of 7075-T6, making a large cost saving over a batch size of 50,000
- Near equal modulus – allowing for an excellent stiffness which can hold up to the required forces.
- Better for casting – due to a reduced copper content
- Better for machining – due to lower hardness, reducing time, wear and cost for any post processing that may be needed (tolerancing holes for example)

When taking into account the batch size, cost and near equal performance, aluminium 6061-T6 is the best material selection for this component.

Manufacturing process

Due to the optimised shape and complex geometries with fillets on multiple planes paired and thin walls, paired with the large batch size of 50,000, high pressure die casting (HPDC) was chosen above other options due to it having the best balance. Aluminium 6061-T6 is able to have a minimum wall thickness of 0.5 mm when used in HPDC. The thinnest wall thickness in the component is $\sim 0.75 \text{ mm}$, making this a suitable process for the component. Due to die casting being chosen, a draft angle was added to the component for ease in ejection from the mold.

Manufacturing constraints reflected in the design

The final design adheres to the following:

- Minimum wall thickness – greater than 0.5 mm
- Draft angle – to ensure ease of ejection
- Filleted radii – minimum 5mm to reduce areas of stress concentration and wear on the die
- Removal of material in low stress areas – to minimise mass

Results

The final crank arm design optimised the strength to weight ratio while remaining in the limits of a maximum Von Mises stress of 220 *MPa* and a maximum deflection of 2 mm.

	Sub optimal part	Final design	Difference (%)
Mass (g)	387.58	190.34	50.89
Max σ_{vm} (MPa)	142.36	219.54	54.21
Max URES deflection (mm)	0.50	1.98	296

FEA of the final design shows it has a maximum Von Mises stress just inside the 220 MPa limit, allowing it to operate at a safe level inside the elastic limit. The deflection of 1.98 mm was just inside the 2 mm limit, allowing a suitable stiffness in the final design. A substantial mass reduction of 50.89% relative to the unoptimized crank has proved the optimisation to be a success in saving material and hence cost. Manufacturing considerations have been taken into account in the construction of the geometry in the evidence of draft angles and a minimum wall thickness within tolerances for the material in HPDC.

Conclusion

The findings of this report show that a substantial 50.89% reduction in crank arm mass can be achieved relative to the sub-optimal part, while still having a maximum Von Mises stress less than 220 MPa and a deflection between 1mm and 2 mm. This demonstrates how optimised load-carrying features that depend on principal stress paths can still maximise the stiffness to weight ratio. This reduction in mass makes use of the full capacity of aluminium 6061-T6's material properties, evidencing how a lightweight component does not inherently result in a weak component. The resulting optimised crank arm, when optimised with a topological optimisation, static study and parametric design can lead to a cheaper component for manufacturing, particularly for a large batch size of 50,000 units. The reduction in mass means double the components can be produced with the same amount of material, reducing overall manufacturing cost when compared to the unoptimized component. Paired with the fact of HPDC being cost effective in large batch quantities, the whole manufacturing process has been optimised, making it a much cheaper part to manufacture.